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Weiqi Wang

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SUMMARY

- Quant-minded applied mathematician (Ph.D.) with 7+ years building algorithms and production-grade research code. Experience spans market data analysis, Python backtesting/paper trading, probabilistic modeling, and forecasting.
- Seeking Quant Research roles where mathematical modeling and robust research engineering drive trading decisions.

SKILLS

Programming	Python (NumPy, pandas, scikit-learn, JAX, PyTorch), kdb+/q
Data/DB	SQL, yfinance, CSV
Tools	Git, Linux, Bash, Jupyter, pytest
Math	Time-series, stochastic processes, optimization, graph algorithms

EDUCATION

Ph.D. in Mathematics, University at Buffalo 2016.08 – 2021.08

Thesis: Numerical Solution to the Corner Angle Problem for an Elastic Solid with Anisotropic Surface Energy

B.S. in Mathematics, Fudan University 2011.09 – 2016.06

PROJECTS

Personal RSI Trading System (solo) 2025.05 – present

- End-to-end **RSI research** and **backtesting** with parameter sweeps and paper-trading support (virtual/actual modes) on **TSX 60** equities.
- Strategies: Base RSI with exit rules, RSI-threshold, analyzers compute total trades, avg return %, take-profit/stop-loss rates, and duration.
- Engineering: designed system architecture; built trading machine with daily data updater and integrity checks; implemented virtual trading system; tests and runbooks.

Baseball Game Outcome Modeling (team project) 2023.09 – 2023.10

- Season-aware simulations and logistic models; **Opponent-Adjusted Efficiency** model performed **better than 538 forecasting model** (Brier score 0.2404 vs 0.2436).

Railway Delay Prediction (Paper on C&IE 2023) 2022.09 – 2023.05

- Cleaned and processed **1M row dataset**; built interpretable **non-homogeneous Markov** forecaster with **chi-square** Markov test and Gaussian-kernel matrix recovery.

Numerical PDEs in Elasticity (Ph.D. work) 2018.01 – 2021.08

- Built high-accuracy solvers (in MATLAB) for anisotropic surface-energy corner-angle problems (boundary-value PDEs); analyzed stability/conditioning and convergence.

SELECTED PUBLICATIONS

- *Approximation schemes for AoI minimization in UAV grid patrols.* Ad Hoc Networks, 2025.
- *A novel Markov model for near-term railway delay prediction.* C & IE, 2023.

EXPERIENCE

Postdoctoral Fellow, University of Victoria 2023.11 – present

- Implemented **JAX** training framework (time-marching PINNs) and reproducible pipelines; evaluation against numerical baselines.

Quant Analyst, NumerixS Quant 2023.06 – 2023.07

- Improved alert precision in daily data integrity checks and added replay tests.

Postdoctoral Fellow, Concordia University 2021.09 – 2023.05

- Developed **numerical methods for high-dimensional PDEs** and approximation algorithms for patrol routing (Age of Information).

This resume only represents the tip of the iceberg of my work, feel free to contact me for more details.